

## Industry view

*In a freewheeling conversation with EFY, Rajesh Arakkal, general manager, FCI India, shares his views on the connector industry in India*

### What is your view on the growth prospects of the connector industry in India?

The connector industry in India is highly diversified and each segment has its own growth pattern. Connectors are required in every industry, from consumer electronics and automotive to industrial production and military and aerospace. Since Make in India, electronic manufacturing clusters and MSIPs are holistic approaches, we expect the demand to rise from all sorts of electronic applications. India is going to become the manufacturing hub of electronic goods in the future, many mobile phone companies are planning to start production facilities in India. Automotive segment is going for an extensive localisation. All these developments add to the growth of the connector industry.

The telecom industry is one of the major contributors and growth drivers for many connectors companies in India through 3G/4G roll-outs. But that does not come overnight. An entire framework comprising spectrum auctions, allocations, partnering with the right technology provider and so on is needed. So, if high-speed technologies like 4G/LTE are picking up the base faster, it will surely be an advantage for the connector industry.

There is a good growth potential for industrial connectors, mainly in the field of automation, power, energy and defence. There will also be potential from the mobile phone manufacturers in India. Domestic demand for industrial connectors in India is expected to grow at least seven per cent to ten per cent in the coming years.

### What are the unique advantages that influence the growth of this sector?

India with over 980 million mobile connections and only nine per cent 3G coverage shows high growth potential.

Growth of telecom connectors depends on the growing 3G/4G roll-outs. This also depends on the decision of telecom equipment manufacturers to make in India at the printed circuit board assembly level.

Increasing demand for fibre-optic connectors, fibre-management systems and cable assembly will be driven by growing data centres, the Internet of Things (IoT) and fibre-to-the-home (FTTH) applications.

The industrial connector demand will be driven by increased



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consumption in power, energy, automotive and automation industries, ATMs, elevators, process automation and so on.

Government procurement for defence, railways, e-governance, etc will generate direct demand for connectors, while government initiatives like Make in India for consumer electronics, mobile phones and more will create demand for domestic component supply chains like connectors.

### What are the hindrances that impact the growth of this sector?

Domestic production of electronics is nowhere close to meeting the increasing demand from India. That is why electronics imports in India exceeded those of precious metals in the first half of 2015. We need to cautiously develop an ecosystem to improve the domestic electronic system design and manufacturing (ESDM) industry.

There is an inverted tax structure in the connector manufacturing that hinders its growth. There is zero duty on import of connectors and terminals, whereas there is five per cent basic duty on raw materials. The government should take immediate actions to stop these irregularities in tax policies.

### Which kind of quality parameters and standards are important for this segment?

Quality expectation of all customers/segments is rising. The customer is driving a zero-defects culture with equal focus on the cosmetic aspect, which, at one time, was focused on only by Japanese customers. Different segments have different quality standards and certifications.

### What is your take on the investment climate in this segment with respect to workforce, investment stability, legal framework, logistics infrastructure and tax conditions?

Investment climate is favourable with the optimism in the market and reduction in the bank rate by RBI. With active implementation of MSIPs, capital investments will happen at a faster rate, which will help the electronic component industry growth.

On the tax side, delay in implementation of GST acts as a hurdle to the smooth flow of material and accumulation of tax credits.

tors can also help reduce weight. As weight and cost savings have become essential for high-reliability products, connector manufacturers are focusing on developing smaller, lighter and higher-power connectors.

### Smaller, tougher ones for wearables

Wearable electronics, whether these are smartwatches, smartglasses, personal health monitors or electronics embedded in clothing, continue to be one of the biggest sectors driving demand for miniature connec-

tors that are able to withstand the rigours of a wide range of environment. These include connectors that can tolerate extremes of temperature, humidity and moisture, while providing tiny form factors that are able to reside within cleverly-packaged electronic devices.

The sector includes devices such as glasses, jewellery, headgear, belts, arm wear, leg wear, skin patches, exoskeletons and e-textiles, with healthcare as the dominant sector for these products, merging medical, fitness and wellness.

The boom in wearable connectors has benefitted the makers of micro-USB connectors, board-to-board connectors, board-to-flexible printed circuit connectors, card connectors, battery connectors and custom cables.

In future, continued emergence of small textiles and e-textiles will challenge connector suppliers as the requirement for washability will increase. Emergence of thin, printable battery technologies may drive the need for a new type of low-profile connector. **EFY**